ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

# ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

## PAPER 1 · PRACTICE D

**MARKS: 100** **TIME: 3 hours**

**NC(V) Programming and Robotics, Level 3. Laid out like the DHET first paper. Section A (30): multiple choice, true or false and matching · Section B (70): DC generators, capacitors, Kirchhoff's laws and series-parallel circuits, cells and batteries, three-phase supply, diode characteristics, oscilloscope readings, logic circuits and soldering. Formula sheet at the end.**

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training examination paper.

## INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B, and a formula sheet.
2. Answer ALL the questions.
3. Read ALL the questions carefully.
4. Number the answers according to the numbering system used in this question paper.
5. Show ALL formulae, substitutions and calculations, and write units with all answers. Round off to TWO decimal places.
6. Draw diagrams neatly, using a ruler. A non-programmable calculator may be used.
7. Write neatly and legibly.

## SECTION A

## QUESTION 1

Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1–1.10) in the ANSWER BOOK.

**1.1** The unit of electrical resistance is the …

A ohm. B volt. C ampere. D watt.

**1.2** Which ONE of the following materials is a good insulator?

A Copper B Aluminium C Rubber D Silver

**1.3** In a DC machine, the part that makes contact with the commutator is the …

A yoke. B brush. C field coil. D shaft.

**1.4** Capacitance is measured in …

A henry. B ohm. C tesla. D farad.

**1.5** The forward voltage drop across a conducting silicon diode is about …

A 0,3 V. B 0,7 V. C 1,5 V. D 5 V.

**1.6** Which gate gives an output of 1 only when all its inputs are 0?

A NOR B NAND C AND D EX-OR

**1.7** Changing alternating current into direct current is called …

A amplification. B modulation.
C induction. D rectification.

**1.8** Kirchhoff's current law states that the currents entering a junction …

A always add up to zero. B are multiplied.
C are equal to the currents leaving it. D are doubled.

**1.9** The time taken for one complete cycle of a waveform is its …

A period. B amplitude. C frequency. D phase.

**1.10** The line voltage of the three-phase supply used in South African factories is about …

A 230 V. B 400 V. C 110 V. D 12 V.

(10)

**[10]**

## QUESTION 2

Indicate whether the following statements are TRUE or FALSE. Write only 'True' or 'False' next to the question number (2.1–2.10) in the ANSWER BOOK.

**2.1** An ammeter is connected in series with the load.

**2.2** The resistance of a conductor is halved when its length is doubled.

**2.3** A transformer works on a steady DC supply.

**2.4** Silicon is the most widely used semiconductor material.

**2.5** An OR gate gives an output of 1 when at least one of its inputs is 1.

**2.6** A fuse protects a circuit by melting when the current becomes too high.

**2.7** Primary cells can be recharged many times.

**2.8** The commutator of a DC motor reverses the current in the armature coil every half turn.

**2.9** A higher frequency means a longer period.

**2.10** The arrow on the emitter of an NPN transistor symbol points outwards.

(10)

**[10]**

## QUESTION 3

Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–M) next to the question number (3.1–3.10) in the ANSWER BOOK.

| | **COLUMN A** | | **COLUMN B** |
|---|---|---|---|
| 3.1 | Increases or decreases an AC voltage | A | voltmeter |
| 3.2 | Lets current flow in one direction only | B | dielectric |
| 3.3 | Number of turns multiplied by the current in a coil | C | transformer |
| 3.4 | Instrument connected in parallel with a component | D | NOT |
| 3.5 | The two states of a transistor used as a switch | E | thermistor |
| 3.6 | Resistor whose resistance changes with temperature | F | diode |
| 3.7 | Insulating material between the plates of a capacitor | G | magnetomotive force |
| 3.8 | Gate drawn as a triangle with a small circle at the output | H | permanent magnet |
| 3.9 | Keeps its magnetism after the magnetising force is removed | I | cut-off and saturation |
| 3.10 | Reduces the ripple in a DC power supply | J | filter capacitor |
| | | K | ammeter |

| | | | |
|---|---|---|---|
| | | L | inductor |
| | | M | AND |

(10)

**[10]**

**TOTAL SECTION A: [30]**

## SECTION B

## QUESTION 4

**4.1** Define the following concepts:

4.1.1 Electromagnetic induction (1)

4.1.2 Lenz's law (1)

4.1.3 Armature (1)

4.1.4 Commutator (1)

**4.2** A DC shunt generator supplies a load current of 40 A at a terminal voltage of 220 V. The armature resistance is 0,2 Ω and the field resistance is 110 Ω. Calculate the:

4.2.1 field current (2)

4.2.2 armature current (2)

4.2.3 emf generated in the armature (3)

**4.3** Explain TWO ways to increase the output voltage of a DC generator. (4)

**[15]**

## QUESTION 5

**5.1** Three capacitors of 4 μF, 6 μF and 12 μF are connected in series. Calculate the total capacitance. (3)

**5.2** Calculate the total capacitance when the same three capacitors are connected in parallel. (2)

**5.3** Currents of 5 A and 3 A flow into a junction. A current of 2 A leaves through one branch. Calculate the current leaving through the only other branch. (2)

**5.4** A 12 V battery is connected to a 4 Ω resistor (R1) in series with a parallel group of a 6 Ω resistor (R2) and a 3 Ω resistor (R3). Calculate the:

5.4.1 total resistance (3)

5.4.2 total current (2)

5.4.3 voltage across R1 (2)

5.4.4 current through R2 (2)

**5.5** State Kirchhoff's voltage law. (2)

**5.6** Give TWO uses of capacitors in the circuits of a robot. (2)

**[20]**

## QUESTION 6

**6.1** Compare primary cells and secondary cells, and give ONE example of each. (4)

**6.2** A 12 V battery rated at 60 Ah supplies a steady current of 4 A. Calculate how long it can supply this current. (2)

**6.3** A star-connected three-phase supply has a line voltage of 400 V. Calculate the phase voltage. (3)

**6.4** Give TWO advantages of a three-phase supply over a single-phase supply. (2)

**6.5** Draw the characteristic curve of a silicon diode. Show the forward and reverse regions, the knee voltage and the breakdown voltage. (4)

**[15]**

## QUESTION 7

**7.1** An oscilloscope is set to 2 V/div and 5 ms/div. A sine wave on the screen is 6 divisions high from its lowest to its highest point, and one complete cycle is 4 divisions wide. Calculate the:

7.1.1 peak-to-peak voltage (2)

7.1.2 peak voltage (1)

7.1.3 period (2)

7.1.4 frequency (2)

**7.2** Inputs A and B are connected to a two-input OR gate. The output of the OR gate and input C are connected to a two-input AND gate, whose output is X. Write the Boolean expression for X and complete a truth table for all combinations of A, B and C. (6)

**7.3** Give TWO input devices and TWO output devices that can be connected to a PLC. (4)

**7.4** Explain how to make a good solder joint on a printed circuit board. (3)

**[20]**

**TOTAL SECTION B: [70]**

**TOTAL: 100**

**END OF QUESTION PAPER**

## FORMULA SHEET

| | |
|---|---|
| Ohm's law: $V = I \times R$ | Power: $P = V \times I = I^2 \times R$ |
| Series: $R_T = R_1 + R_2 + R_3$ | Parallel: $1 \div R_T = 1 \div R_1 + 1 \div R_2 + 1 \div R_3$ |
| Capacitors in series: $1 \div C_T = 1 \div C_1 + 1 \div C_2 + 1 \div C_3$ | Capacitors in parallel: $C_T = C_1 + C_2 + C_3$ |
| Generator: $E = V + I_aR_a$; $I_a = I_L + I_f$ | Motor: $E = V - I_aR_a$ |
| Star connection: $V_L = \sqrt{3} \times V_{ph}$; $I_L = I_{ph}$ | Delta connection: $V_L = V_{ph}$; $I_L = \sqrt{3} \times I_{ph}$ |
| Battery capacity: $Ah = I \times t$ | Frequency: $f = 1 \div T$ |
| Peak voltage: $V_p = V_{p\text{-}p} \div 2$ | Kirchhoff: $\Sigma I$ in = $\Sigma I$ out; $\Sigma$ voltage rises = $\Sigma$ voltage drops |
| Resistivity: $R = \rho \times L \div A$ | Transformer: $V_p \div V_s = N_p \div N_s$ |